

Lesson 09 Demo 01

Creating Azure SQL DB

Objective: To create an Azure SQL DB to efficiently store and manage data

Tools required: Azure account with administrator access

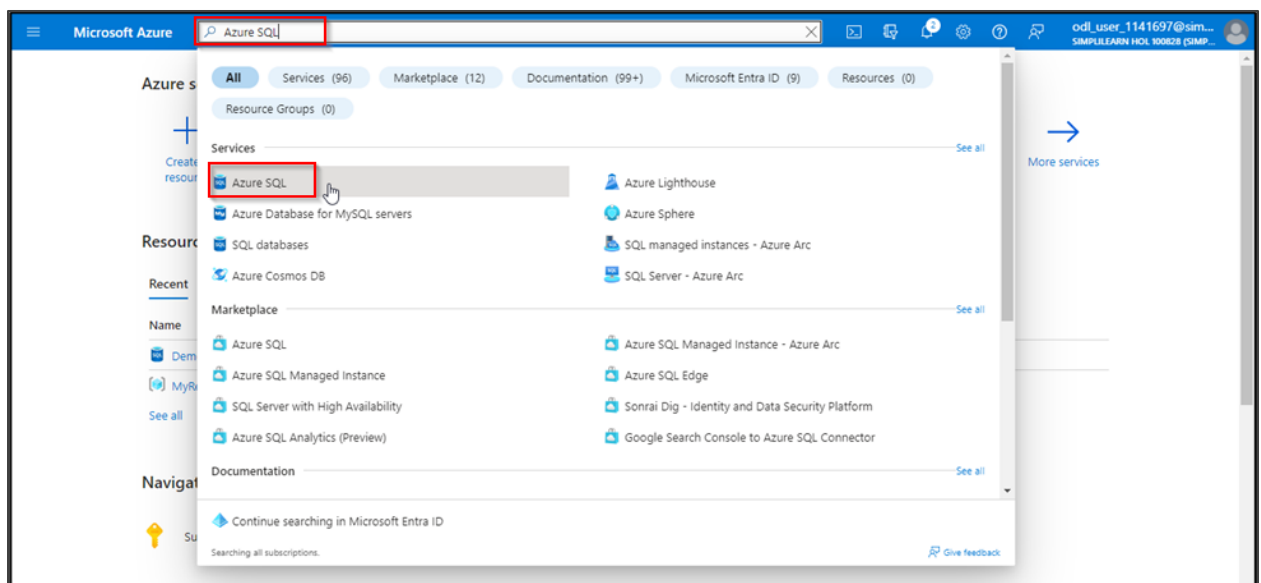
Prerequisites: None

Steps to be followed:

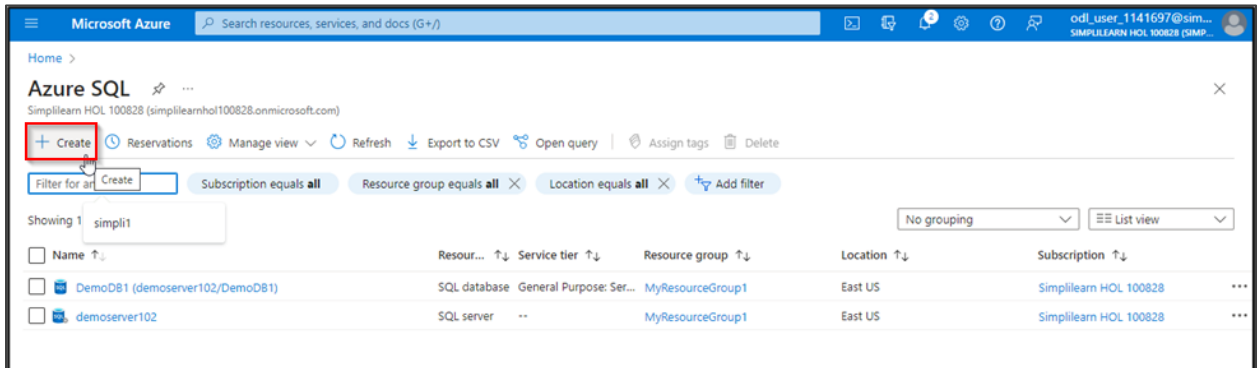
1. Create the SQL database

Step 1: Create the SQL database

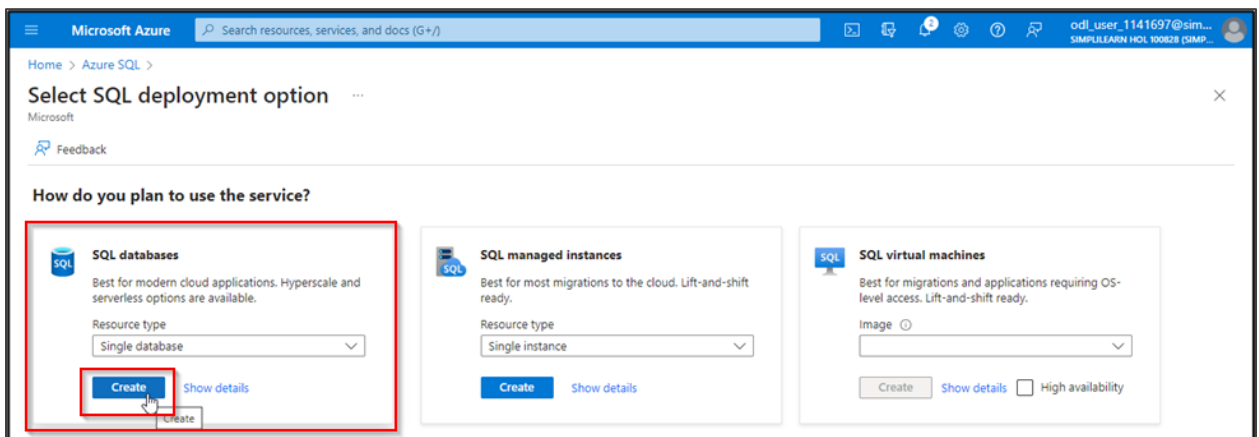
- 1.1 Log in to the Azure portal; search for and select **Azure SQL** in the search box



1.2 In the **Azure SQL** page, click on the **+ Create** button



1.3 In the **Select SQL deployment option** page, select **SQL databases**. Under **Resource type**, choose **Single database** and then click on **Create**



- 1.4 In the **Create SQL Database** page, in the **Subscription** field, choose an available subscription you are working under. For **Resource group**, you can select an available resource group or create a new one by clicking on **Create new**

Microsoft Azure

Search resources, services, and docs (G+)

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Want to try Azure SQL Database for free? Create a free serverless database with the first 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of the subscription. [Learn more](#)

[Apply offer \(Preview\)](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

[Review + create](#) [Next: Networking >](#)

Cost summary

General Purpose (GP_S_Gen5_1)

Cost per GB (in --) --

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH --

COMPUTE COST / VCORE SECOND ¹ --

NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

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- 1.5 Enter the resource group name in **Name** field and click on **OK**

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Home > Azure SQL > Select SQL deployment option >

Create SQL Database

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[Review + create](#) [Next: Networking >](#)

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A resource group is a container that holds related resources for an Azure solution.

Name *

[OK](#) [Cancel](#)

1.6 Enter a name in the **Database name** field, and for the **Server**, click on **Create new**

Microsoft Azure Search resources, services, and docs (G+)

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *

Server * [Create new](#)

☒ The value must not be empty.

Want to use SQL elastic pool? ☐ Yes ☒ No

Workload environment ☒ Development

[Review + create](#) [Next: Networking >](#)

ESTIMATED STORAGE COST / MONTH

COMPUTE COST / VCORE SECOND ¹

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- ✓ Database name should not match special patterns
- ✓ The value has a length of at most 128.
- ✓ Database name should not contain reserved words
- ✓ No database with the same name exists in the server

1.7 In the **Create SQL Database Server** page, enter **Server name** and select **Authentication method** as **Use SQL authentication**

Microsoft Azure

Home > Azure SQL > Select SQL deployment option > Create SQL Database

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * .database.windows.net

Location *

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☒ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Set Microsoft Entra admin *

Not Selected

[Set admin](#)

OK

1.8 Enter the remaining required details as shown in the screenshot below and then click on the **OK** button:

Microsoft Azure

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Home > Azure SQL > Select SQL deployment option > Create SQL Database >

Create SQL Database Server

Microsoft

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * demoserver103 ✓
.database.windows.net

Location * (US) East US ▼

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login * Demo_DB2 ✓

Password * ✓

Confirm password * ✓

OK

1.9 In the **Create SQL Database** page, click on **Configure database** under **Compute + storage**

Microsoft Azure

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Want to use SQL elastic pool? ☐ Yes ☒ No

Workload environment ☒ Development ☐ Production

Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage * ☐ **General Purpose - Serverless**
Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled
Configure database

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy ☐ Locally-redundant backup storage ☐ Zone-redundant backup storage ☒ Geo-redundant backup storage

Warning: Selected value for backup storage redundancy is Geo-redundant backup storage. Database backups will be geo-replicated which might impact your data residency requirements. [Learn more](#)

1.10 In the **Configure** page, select the configurations per the requirement and click on the **Apply** button

Microsoft Azure

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Home > Azure SQL > Select SQL deployment option >

Configure

Feedback

up to 80 vCores, up to 240 GB memory
[Change configuration](#)

NOTES
1 Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

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Max vCores: 2

Min vCores: 0.5 vCores

2.05 GB MIN MEMORY 6 GB MAX MEMORY

Auto-pause delay
The database automatically pauses if it is inactive for the time period specified here, and automatically resumes when database activity recurs. Alternatively, auto-pausing can be disabled.

☒ Enable auto-pause

Days: 0 Hours: 1 Minutes: 0

Data max size (GB): 32

9.6 GB LOG SPACE ALLOCATED

Would you like to make this database zone redundant? ☐ Yes ☒ No

Apply

- 1.11 Select **Locally-redundant backup storage** under **Backup storage redundancy** and then click on **Next : Networking >**

Microsoft Azure

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Database name * DemoDB2 ✓

Server * ⓘ (new) demoserver103 (East US) [Create new](#)

Want to use SQL elastic pool? ⓘ ☐ Yes ☒ No

Workload environment ☒ Development ☐ Production

i Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage * ⓘ **General Purpose - Serverless**
Standard-series (Gen5), 2 vCores, 32 GB storage, zone redundant disabled
[Configure database](#)

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy ⓘ ☒ Locally-redundant backup storage
☐ Zone-redundant backup storage
☐ Geo-redundant backup storage

[Review + create](#) [Next : Networking >](#)

1.12 In the **Networking** tab, select the options as shown in the screenshot below and then go to the **Additional settings** tab:

Microsoft Azure

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Basics **Networking** Security **Additional settings** Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'demoserver103' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method *

☐ No access

☒ **Public endpoint**

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

Add current client IP address *

Connection policy

Configure how clients communicate with your SQL database server. [Learn more](#)

Connection policy *

☒ **Default - Uses Redirect policy for all client connections originating inside of Azure (except Private Endpoint connections) and Proxy for all client connections originating outside Azure**

SQL

Cost summary

General Purpose (GP S Gen5 2)

Cost per GB (in --)

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH -- ..

COMPUTE COST / VCORE SECOND -- ..

NOTES

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1.13 In the **Additional settings** tab, select **Sample** under **Use existing data** field and then click on **Review + create** button

Microsoft Azure

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Basics Networking Security **Additional settings** Tags Review + create

Customize additional configuration parameters including collation & sample data.

Data source

Start with a blank database, restore from a backup or select sample data to populate your new database.

Use existing data * None Backup **Sample**

AdventureWorksLT will be created as the sample database.

Database collation

Database collation defines the rules that sort and compare data, and cannot be changed after database creation. The default database collation is SQL_Latin1_General_CP1_CI_AS. [Learn more](#)

Collation ⓘ

Cost summary

General Purpose (GP 5 Gen5 2)

Cost per GB (in --) --

Max storage selected (in GB) x **41.6**

ESTIMATED STORAGE COST / MONTH --

COMPUTE COST / VCORE SECOND ¹ --

NOTES

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Review + create < Previous Next: Tags >

1.14 In the **Review + create** tab, click on the **Create** button

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Home > Azure SQL > Select SQL deployment option >

Create SQL Database ...

Microsoft

Basics

Networking

Security

Additional settings

Tags

Review + create

Product details

SQL database

by Microsoft

Terms of use | Privacy policy

Estimated cost

Storage cost -- -- / month + Compute cost -- -- / vCore second

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription

Simplilearn HOL 100828

Resource group

MyResourceGroup2

Region

East US

Database name

DemoDB2

Server

(new) demoserver103

Authentication method

SQL authentication

Server admin login

Demo_DB2

Compute + storage

General Purpose - Serverless: Standard-series (Gen5), 2 vCores, 32 GB storage, zone redundant disabled

Backup storage redundancy

Locally-redundant backup storage

SQL

Cost summary

General Purpose (GP, 5 Gen5 2)

Cost per GB (in --)

--

Max storage selected (in GB)

x 41.6

ESTIMATED STORAGE COST / MONTH

-- --

COMPUTE COST / VCORE SECOND¹

-- --

NOTES

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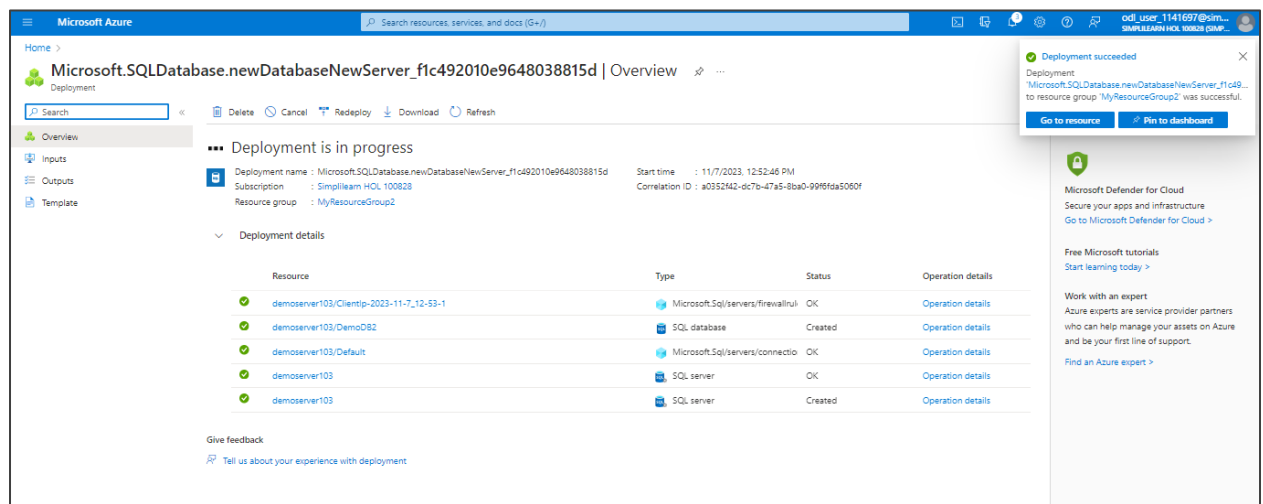
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Create

< Previous

[Download a template for automation](#)

After successful deployment, the following screen will appear:



Microsoft Azure

Home > Microsoft.SQLDatabase.newDatabaseNewServer_f1c492010e9648038815d | Overview

Deployment

Search resources, services, and docs (G+)

Deployment is in progress

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_f1c492010e9648038815d
Subscription : Simplilearn HCL 100828
Resource group : MyResourceGroup2

Start time : 11/7/2023, 12:52:46 PM
Correlation ID : a0352442-6c7b-47a5-8ba0-99f6fda5060f

Deployment details

Resource	Type	Status	Operation details
demosever103/Clientip-2023-11-7_12-53-1	Microsoft.Sql/servers/firewallrul	OK	Operation details
demosever103/DemoDB2	SQL database	Created	Operation details
demosever103/Default	Microsoft.Sql/servers/connectio	OK	Operation details
demosever103	SQL server	OK	Operation details
demosever103	SQL server	Created	Operation details

Give feedback
Tell us about your experience with deployment

Deployment succeeded

Deployment: "Microsoft.SQLDatabase.newDatabaseNewServer_f1c492010e9648038815d" to resource group "MyResourceGroup2" was successful.

Go to resource Pin to dashboard

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By following the above steps, you have successfully created the Azure SQL DB to store and manage data efficiently.

Note: If the deployment is unsuccessful, navigate to policy definition and disable the policy enforcement.